Self-Study Week

8051 Microcontroller, Chapter 6: Arithmetic and Logic Instructions

1. Instructions to students

This week there are no lectures for Chapter 6. You will study the chapter yourself, using the chapter notes, and then test your understanding with the exercises in Section 2. The exercises are a self-check: they show you what you understand and what you must review.

2. Exercises

Part A: ADD and ADDC

A1. *(2 marks)* Give the final values of A (in hex), CY, and AC after this code.

```
CLR   C
MOV   A, #0F5H
ADDC  A, #0BH
```

A2. *(2 marks)* Give the final values of A (in hex), CY, and AC after this code.

```
SETB  C
MOV   A, #3AH
ADDC  A, #26H
```

A3. *(3 marks)* Write a program that adds the 16-bit numbers 12B7H and 0E69H. Save the result in R6 (low byte) and R7 (high byte). State the final values of R7 and R6.

A4. *(1 mark)* A student runs MOV A,#7 followed by ADDC A,#2 in two different programs. One program gives A = 09H and the other gives A = 0AH. Explain why, and how to make the result always 09H.

Part B: SUBB

B1. *(2 marks)* Give the final values of A (in hex), CY, and AC after this code.

```
MOV   A, #52H
CLR   C
SUBB  A, #2FH
```

B2. *(3 marks)* Give the final value of A and CY after this code. Is the result positive or negative? Find the size of the result.

```
MOV   A, #20H
CLR   C
SUBB  A, #30H
```

B3. *(1 mark)* Give the final value of A after this code.

```
SETB  C
MOV   A, #09H
SUBB  A, #04H
```

B4. *(1 mark)* List the three steps the CPU performs to execute SUBB.

Part C: Packed BCD, unpacked BCD, and DA A

C1. *(2 marks)* Write the decimal number 47 as packed BCD and as unpacked BCD (in hex).

C2. *(2 marks)* Which of these bytes are valid packed BCD: 59H, 7AH, 90H, 1FH? Give the reason for each invalid one.

C3. *(2 marks)* Give the value of A after ADD and after DA A.

```
MOV   A, #38H
ADD   A, #45H
DA    A
```

C4. *(2 marks)* Give the value of A and AC after ADD, and the value of A after DA A. Explain why DA A changes the result even though the low nibble looks valid.

```
MOV   A, #38H
ADD   A, #29H
DA    A
```

C5. *(2 marks)* Give the value of A after ADD and the final values of A and CY after DA A.

```
MOV   A, #86H
ADD   A, #47H
DA    A
```

C6. *(2 marks)* Give the value of A after ADD, and the final A and CY after DA A. Explain why DA A also adds 60H even though the upper nibble of the ADD result is 9.

```
MOV   A, #49H
ADD   A, #51H
DA    A
```

C7. *(1 mark)* True or false: DA A can be used after SUBB to correct a packed BCD subtraction. Give the reason.

C8. *(3 marks)* Write a program that adds the packed BCD numbers 76 and 58. Save the tens and ones digits in R6 and the hundreds digit in R7. State the final values of R7 and R6.

Part D: MUL and DIV

D1. *(2 marks)* Give the final values of A, B, and OV (in hex where needed).

```
MOV   A, #0F0H
MOV   B, #10H
MUL   AB
```

D2. *(1 mark)* Give the final values of A, B, and OV.

```
MOV   A, #12H
MOV   B, #05H
MUL   AB
```

D3. *(2 marks)* Give the final values of A and B in hex.

```
MOV   A, #200
MOV   B, #7
DIV   AB
```

D4. *(1 mark)* What does OV = 1 after DIV AB tell you?

D5. *(2 marks)* Write a program that uses DIV to test the number in R1: put 00H in R0 if the number is even and 01H if it is odd.

D6. *(2 marks)* R3 contains 9BH. Give the values of R5, R6, and R7 after this program (hex to decimal digits).

```
MOV   A, R3
MOV   B, #10
DIV   AB
MOV   R5, B
MOV   B, #10
DIV   AB
MOV   R6, B
MOV   R7, A
```

D7. *(2 marks)* R2 = 07H (tens digit) and R3 = 04H (ones digit). Give the final value of A in hex, and the decimal number it represents.

```
MOV   A, R2
MOV   B, #10
MUL   AB
ADD   A, R3
```

Part E: ANL and ORL

E1. *(1 mark)* Give the final value of A.

```
MOV   A, #0B7H
ANL   A, #0F0H
```

E2. *(1 mark)* Write one instruction that clears bits 3 and 2 of A and leaves all other bits unchanged.

E3. *(1 mark)* Write one instruction that sets bits 7 and 0 of A and leaves all other bits unchanged.

E4. *(2 marks)* (a) A contains 37H, the ASCII code of the character "7". Write one instruction that converts it to unpacked BCD. (b) A contains 09H (unpacked BCD). Write one instruction that converts it to ASCII.

Part F: Integrated task

F1. *(4 marks)* R2 = 34H and R3 = 36H are the ASCII codes of the digits "4" and "6". Write a program that converts them into one binary number in A (the number 46 decimal). Use ANL, MUL, and ADD. State the final value of A in hex.

3. Evaluate your study

Compare your answers with the answer key when your instructor gives it to you. Give yourself the marks for each part.

Part	Topic	Marks available	Your marks
Part A	ADD and ADDC	8	
Part B	SUBB	7	
Part C	Packed BCD, unpacked BCD, and DA A	16	

Part	Topic	Marks available	Your marks
Part D	MUL and DIV	12	
Part E	ANL and ORL	5	
Part F	Integrated task	4	
Total		52	

Your total	What to do
42 to 52 (80% or more)	You are ready for the next topic. Review only the questions you lost marks on.
32 to 41 (60% to 79%)	Re-read the topics where you lost marks, redo those questions without notes, and ask your instructor about anything still unclear.
Below 32 (under 60%)	Study the chapter notes again, topic by topic, and contact your instructor before the next class.